Experts, Errors, and Context: A Large-Scale Study of Human Evaluation for Machine Translation

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INTRODUCTION

Problem:

Although modern machine translation (MT) models have achieved high fluency, deep semantic errors remain common. As a result, accurately measuring true translation quality has become increasingly difficult.

Motivation:

The main motivation of this study is to question how reliably existing evaluation methods rank modern MT systems, and to propose a context-aware, error-based human evaluation standard that more accurately reflects true translation quality.

WHAT IS THE AIM?

- To identify under which conditions different human evaluation approaches (MQM, SQM, WMT) are reliable or unreliable.
- To examine how consistent these methods are in system-level and segment-level rankings.
- To determine which evaluation method best distinguishes true quality differences between MT systems.
- To analyze the gap between document-level and segment-level evaluations.

WHAT IS THE METHOD?

- Different MT systems are evaluated.
- The experiments are conducted on the WMT 2020 English→German and Chinese→English test sets.
- Three human evaluation methods are used: MQM, SQM, and WMT.
- Both segment-level and document-level scores are compared.
- Correlation analyses (Pearson, Kendall, Spearman) are used to measure the reliability of the evaluation metrics.

HUMAN EVALUATION METHODOLOGIES

WMT

SQM

MQM

WMT

- WMT (Workshop on Machine Translation) is an international evaluation platform designed to compare machine translation systems.
- The official human evaluation method used by WMT is crowd-sourced Direct Assessment (DA)
- Annotators score each segment on a 0–100 scale.
- Evaluation is performed at the segment level; no document-level context is provided.
- Annotators are typically non-expert crowd workers.
- In segment-level evaluation, WMT produces two types of scores: a raw score and a z-normalized score adjusted per annotator; document-level and system-level scores are computed as the average of these segment scores.

SQM

- SQM is a human evaluation method in which translation segments are scored on a 0–6 scale.
- Evaluations are performed by professional translators rather than crowd workers.(Crowd workers can also be used in simplified versions e.g. csqm)
- The rater can view surrounding sentences in the document while judging a segment.
- Each segment is scored independently; raters do not annotate error types.
- A system's overall SQM score is computed as the average of all segment scores.

MQM

MQM is a human evaluation method that measures translation quality using detailed error types and weights.

Raters mark each error in a segment based on its category and severity, and the method is fully context-aware. In this study, MQM is used as a customized version of the MQM Core hierarchy.

MQM

- Raters identify specific errors rather than assigning a single score.
- Raters make use of full document context.
- A maximum of five errors may be marked per segment.
- “Non-translation” is included as a dedicated error category.
- Each error is annotated with a category and a severity level.
- Errors are weighted, and a segment-level score is produced accordingly.
- The system-level score is computed as the average of all segment scores.
- Raters are selected from professional translators.
- Minor punctuation errors receive a dedicated weight.

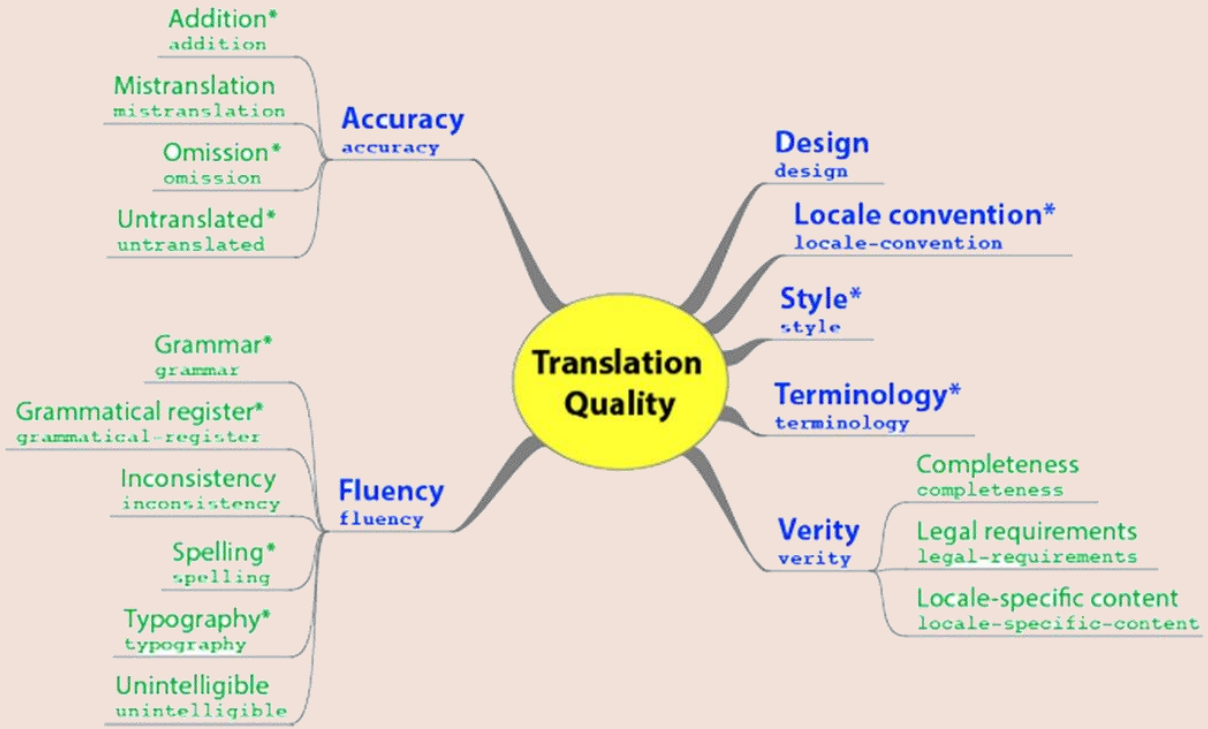


Figure 10: MQM Core issue hierarchy.

This study uses an additional dedicated category called *Non-translation* on top of the standard MQM Core hierarchy.

In MQM, each error is first classified according to its severity level, and the final segment score is computed using the corresponding severity weights. These tables present the customized severity definitions and error weights used in the study.

Severity	Description
Major	Errors that may confuse or mislead the reader due to significant change in meaning or because they appear in a visible or important part of the content.
Minor	Errors that don't lead to loss of meaning and wouldn't confuse or mislead the reader but would be noticed, would decrease stylistic quality, fluency or clarity, or would make the content less appealing.
Neutral	Use to log additional information, problems or changes to be made that don't count as errors, e.g. they reflect a reviewer's choice or preferred style.

Table 11: MQM severity levels.

Severity	Category	Weight
Major	Non-translation	25
	all others	5
Minor	Fluency/Punctuation	0.1
	all others	1
Neutral	all	0

Table 1: MQM error weighting.

Experimental Setup

System	WMT↑	WMT RAW↑	cSQM↑	pSQM↑	MQM ↓	Major↓	Minor↓	Fluency↓	Accuracy↓
Human-B	0.569(1)	90.5(1)	5.31(1)	5.16(1)	0.75(1)	0.22(1)	0.54(1)	0.28(1)	0.47(1)
Human-A	0.446(4)	85.7(4)	5.20(2)	4.90(2)	0.91(2)	0.28(2)	0.64(2)	0.33(2)	0.58(2)
Human-P	0.299(10)	84.2(9)	5.04(5)	4.32(3)	1.41(3)	0.57(3)	0.85(3)	0.50(3)	0.91(3)
Tohoku-AIP-NTT	0.468(3)	88.6(2)	5.11(3)	3.95(4)	2.02(4)	0.94(4)	1.14(4)	0.61(5)	1.40(4)
OPPO	0.495(2)	87.4(3)	5.03(6)	3.79(5)	2.25(5)	1.07(5)	1.19(6)	0.62(6)	1.63(5)
eTranslation	0.312(9)	82.5(10)	5.02(7)	3.68(7)	2.33(6)	1.18(7)	1.16(5)	0.56(4)	1.78(7)
Tencent_Translation	0.386(6)	84.3(8)	5.06(4)	3.77(6)	2.35(7)	1.15(6)	1.22(8)	0.63(7)	1.73(6)
Huoshan_Translate	0.326(7)	84.6(6)	5.00(8)	3.65(8)	2.45(8)	1.23(8)	1.23(9)	0.64(8)	1.80(8)
Online-B	0.416(5)	84.5(7)	4.95(9)	3.60(9)	2.48(9)	1.34(9)	1.20(7)	0.64(9)	1.84(9)
Online-A	0.322(8)	85.3(5)	4.85(10)	3.32(10)	2.99(10)	1.73(10)	1.32(10)	0.76(10)	2.23(10)

Table 3: English→German: Different human evaluations for 10 submissions of the WMT20 evaluation campaign.

System	WMT↑	WMT RAW↑	cSQM↑	pSQM↑	MQM ↓	Major↓	Minor↓	Fluency↓	Accuracy↓
Human-A	-	-	5.09(2)	4.34(1)	3.43(1)	2.71(1)	0.74(1)	0.91(1)	2.52(1)
Human-B	-0.029(9)	74.8(9)	5.03(7)	4.29(2)	3.62(2)	2.81(2)	0.82(10)	0.95(2)	2.66(2)
VolcTrans	0.102(1)	77.47(5)	5.04(5)	4.03(3)	5.03(3)	4.26(3)	0.79(6)	1.31(7)	3.71(3)
WeChat_AI	0.077(3)	77.35(6)	4.99(8)	4.02(4)	5.13(4)	4.39(4)	0.76(4)	1.24(5)	3.89(4)
Tencent_Translation	0.063(4)	76.67(7)	5.04(6)	3.99(5)	5.19(5)	4.43(6)	0.79(8)	1.23(4)	3.96(5)
OPPO	0.051(7)	77.51(4)	5.07(4)	3.99(5)	5.20(6)	4.41(5)	0.81(9)	1.23(3)	3.97(6)
THUNLP	0.028(8)	76.48(8)	5.11(1)	3.98(7)	5.34(7)	4.61(7)	0.75(3)	1.27(6)	4.07(9)
DeepMind	0.051(6)	77.96(1)	5.07(3)	3.97(8)	5.41(8)	4.67(8)	0.75(2)	1.38(8)	4.02(7)
DiDi_NLP	0.089(2)	77.63(3)	4.91(9)	3.95(9)	5.48(9)	4.73(9)	0.77(5)	1.43(9)	4.05(8)
Online-B	0.06(5)	77.77(2)	4.83(10)	3.89(10)	5.85(10)	5.08(10)	0.79(7)	1.51(10)	4.34(10)

Table 4: Chinese→English: Different human evaluations for 10 submissions of the WMT20 evaluation campaign.

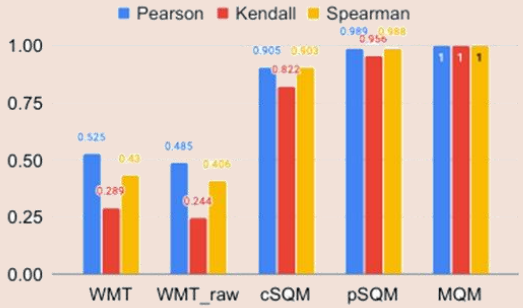


Figure 1: English→German: System correlation with the platinum ratings acquired with MQM.

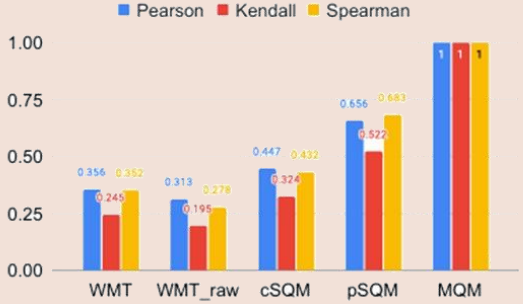


Figure 2: English→German: Segment correlation with the platinum ratings acquired with MQM.

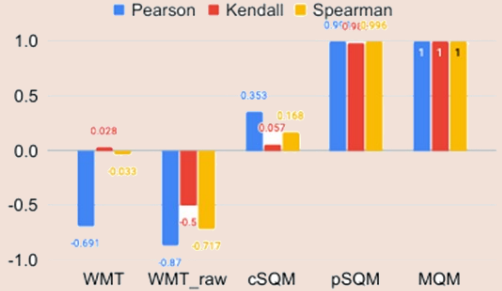


Figure 3: Chinese→English: System-level correlation with the platinum ratings acquired with MQM.

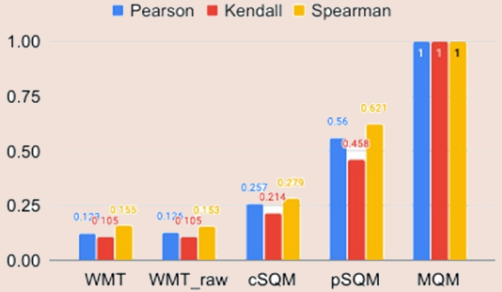


Figure 4: Chinese→English: Segment correlation with the platinum ratings acquired with MQM.

Error Categories	Errors (%)	Major (%)	Human MQM	All MT MQM	All MT vs H.	Tohoku MQM	Tohoku vs H.	OPPO MQM	OPPO vs H.	eTrans MQM	eTrans vs H.
Accuracy/Mistranslation	33.2	27.2	0.296	1.285	*4.3*	1.026	*3.5*	1.219	*4.1*	1.244	*4.2*
Style/Awkward	14.6	4.6	0.146	0.299	*2.0*	0.289	*2.0*	0.315	*2.1*	0.296	*2.0*
Fluency/Grammar	10.7	4.7	0.097	0.224	*2.3*	0.193	*2.0*	0.215	*2.2*	0.196	*2.0*
Accuracy/Omission	3.6	13.4	0.070	0.091	*1.3*	0.063	*0.9*	0.063	*0.9*	0.120	*1.7*
Accuracy/Addition	1.8	6.7	0.067	0.025	*0.4*	0.018	*0.3*	0.024	*0.4*	0.021	*0.3*
Terminology/Inappropriate	8.3	7.0	0.061	0.193	*3.2*	0.171	*2.8*	0.189	*3.1*	0.193	*3.2*
Fluency/Spelling	2.3	1.2	0.030	0.039	*1.3*	0.030	*1.0*	0.039	*1.3*	0.028	*0.9*
Accuracy/Untranslated tex	3.1	14.9	0.024	0.090	*3.8*	0.082	*3.5*	0.066	*2.8*	0.098	*4.2*
Fluency/Punctuation	20.3	0.2	0.014	0.039	*2.8*	0.067	*4.9*	0.013	*1.0*	0.011	*0.8*
Other	0.5	5.2	0.005	0.010	*1.9*	0.009	*1.6*	0.010	*1.9*	0.007	*1.2*
Fluency/Register	0.6	5.0	0.005	0.014	*3.0*	0.009	*1.9*	0.015	*3.2*	0.015	*3.3*
Terminology/Inconsistent	0.3	0.0	0.004	0.005	*1.2*	0.004	*0.9*	0.005	*1.2*	0.005	*1.2*
Non-translation	0.2	100.0	0.003	0.083	*28.3*	0.041	*14.0*	0.065	*22.0*	0.094	*32.0*
Fluency/Inconsistency	0.1	1.3	0.003	0.002	*0.7*	0.001	*0.3*	0.001	*0.3*	0.003	*1.0*
Fluency/Character enc.	0.1	3.7	0.002	0.001	*0.7*	0.002	*1.0*	0.001	*0.6*	0.000	*0.2*
All accuracy	41.7	24.2	0.457	1.492	*3.3*	1.189	*2.6*	1.372	*3.0*	1.483	*3.2*
All fluency	34.2	1.8	0.150	0.320	*2.1*	0.303	*2.0*	0.284	*1.9*	0.253	*1.7*
All except acc. & fluenc	24.2	6.0	0.222	0.596	*2.7*	0.526	*2.4*	0.591	*2.7*	0.596	*2.7*
All categories	100.0	12.1	0.829	2.408	*2.9*	2.017	*2.4*	2.247	*2.7*	2.332	*2.8*

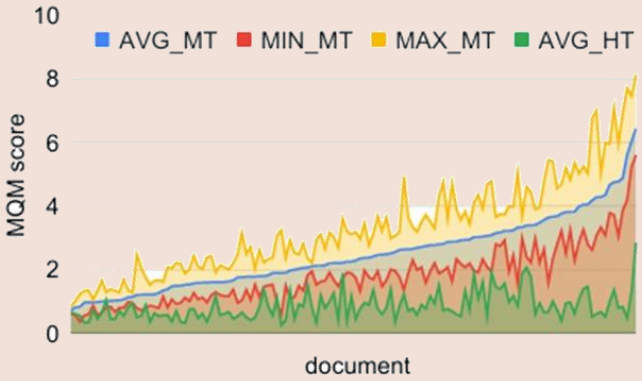


Figure 5: EnDe: Document-level MQM scores.

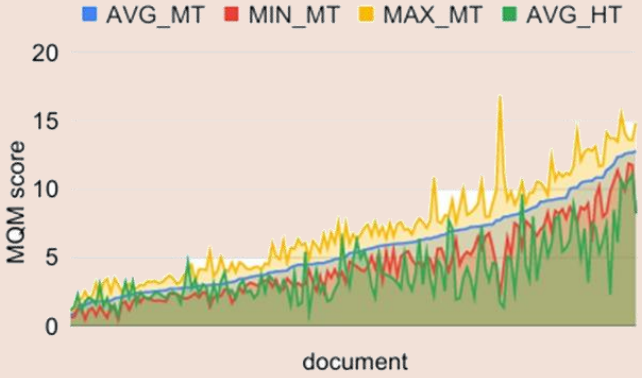
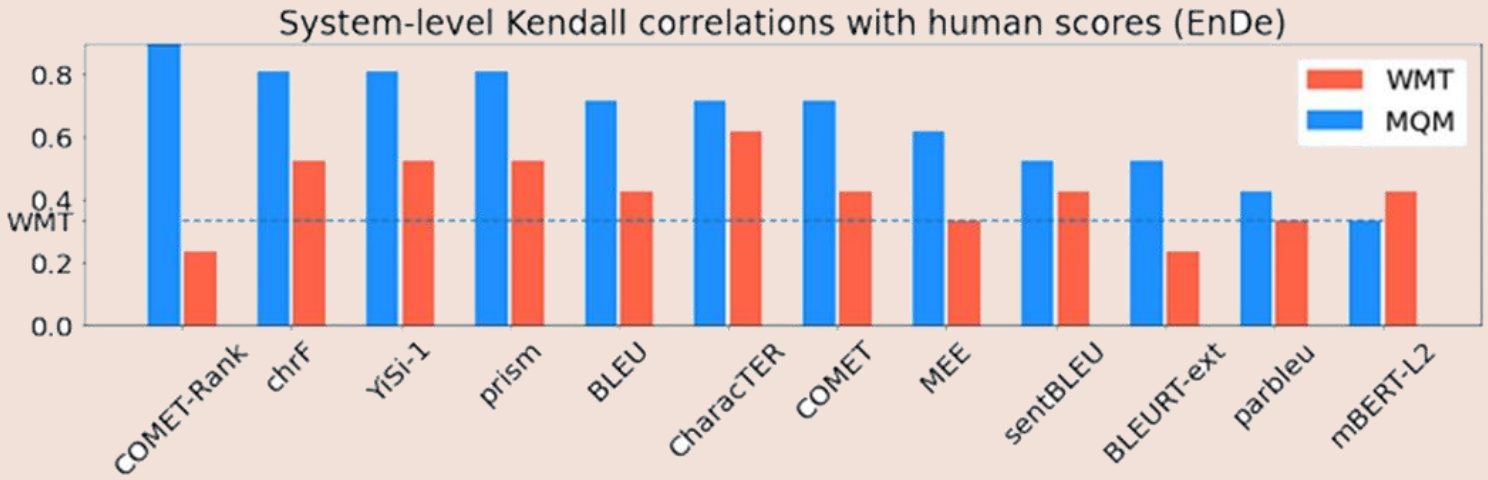
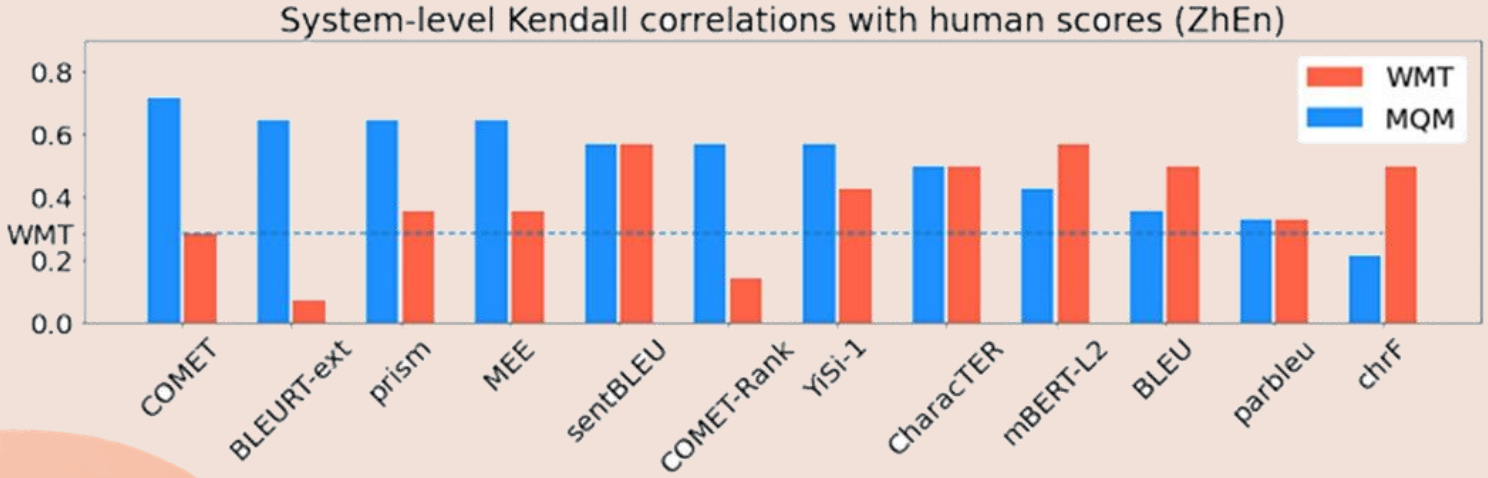


Figure 6: ZhEn: Document-level MQM scores.


System-level Kendall correlations with human scores (EnDe)
0.8
0.6
0.4
WMT
0.2
0.0
WMT
MQM
COMET-Rank
chrF
YiSi-1
prism
BLEU
CharacTER
COMET
MEE
sentBLEU
BLEURT-ext
parbleu
mBERT-L2


System-level Kendall correlations with human scores (ZhEn)
0.8
0.6
0.4
WMT
0.2
0.0
WMT
MQM
COMET
BLEURT-ext
prism
MEE
sentBLEU
COMET-Rank
YiSi-1
CharacTER
mBERT-L2
BLEU
parbleu
chrF

Average correlations	EnDe		ZhEn	
	WMT	MQM	WMT	MQM
Pearson, sys-level	0.539	0.883	0.318	0.551
	0.23	*0.02*	*0.41*	*0.21*
Kendall, sys-level	0.436	0.637	0.309	0.443
	0.27	*0.10*	*0.42*	*0.23*
Kendall, sys-level,	0.467	0.676	0.514	0.343
baseline metrics	*0.20*	*0.06*	*0.10*	*0.34*
Kendall, sys-level,	0.387	0.123	0.426	0.159
+ human	*0.26*	*0.68*	*0.20*	*0.64*
Kendall, seg-level	0.170	0.228	0.159	0.298
	0.00	*0.00*	*0.00*	*0.00*
Kendall, seg-level,	0.159	0.161	0.157	0.276
+ human	*0.00*	*0.00*	*0.00*	*0.00*

Table 9: Average correlations for various subsets of metrics at different granularities. Numbers in italics are average p-values from two-tailed tests, indicating the probability that the observed correlation was due to chance.

Conclusion

- Inconsistencies appear in system rankings across MQM, SQM, and WMT, and crowd-based evaluations in particular are noisy and less reliable.
- MQM's context-aware structure produces the most stable rankings.
- Accuracy-related errors remain the major challenge, while fluency is largely solved.
- Document-level evaluation yields results that differ substantially from segment-level evaluation.
- MQM is costly but the most reliable; SQM provides a practical intermediate solution.
- The Non-translation category is critical for properly capturing the impact of severely poor outputs.
- Correlation measures such as Pearson, Kendall, and Spearman capture different properties of metric reliability, indicating that no single correlation metric is sufficient on its own.
- COMET shows the closest system-level correlation to MQM, whereas BLEU fails to capture semantic errors.
- No automatic metric reaches MQM's level of granularity at the segment level, demonstrating that automatic metrics cannot fully replace human evaluation.

THANKS

Do you have any questions?